

Supplement to Six Birds Foundations VII: Lawful Theory Interaction

Ioannis Tsiokos 

Version 2: 6 October 2026
(Version 1: 2 September 2026)

S1 What this supplement holds

This supplement accompanies *Six Birds Foundations VII: Lawful Theory Interaction*. The main paper states and proves every result; its numbered items are cited here by their numbers there. The supplement gives the Lean counterpart of each numbered item (Supplement Section S2), the field lists of the certificates (Supplement Section S5), the register of the 36 catalog entries with their refusals and cross-family corollaries (Supplement Section S6), the finite evidence and the named countermodels (Supplement Section S7), the formal status of the Lean development (Supplement Section S8) and the commands that rebuild it (Supplement Section S9). The certificate language builds on the series' earlier papers [3, 4, 5, 6, 7, 8].

Every declaration listed is present in the Lean sources at the file and line given.

S2 Source concordance

Table S1 lists every numbered item of the main paper with its catalog identifier, where it has one, and its Lean declarations with their kind, file and line. Items marked *new* have no catalog identifier; their declarations are in the semantic modules, except where the row states that the item is argued on paper only. Items marked *semantic* give the semantic counterpart of the catalog entry named.

Table S1. Source concordance.

Item	Catalog	Lean declarations
Definition 4.1	new	<code>Semantics.TheoryPackage</code> (structure) <code>formalization/lean/FoundationsVII/Semantics/Join.lean:7</code> <code>Semantics.SemanticJoin</code> (structure) <code>formalization/lean/FoundationsVII/Semantics/Join.lean:11</code> <code>Semantics.PackageJoin</code> (structure) <code>formalization/lean/FoundationsVII/Semantics/Join.lean:15</code>
Definition 4.2	new	<code>Semantics.Factors</code> (def) <code>formalization/lean/FoundationsVII/Semantics/Join.lean:21</code> <code>Semantics.FactorsOnImage</code> (def) <code>formalization/lean/FoundationsVII/Semantics/Join.lean:26</code> <code>Semantics.SplitPair</code> (def) <code>formalization/lean/FoundationsVII/Semantics/Join.lean:36</code> <code>Semantics.factors_implies_onImage</code> (theorem) <code>formalization/lean/FoundationsVII/Semantics/Join.lean:31</code>

Item	Catalog	Lean declarations
Theorem 4.3	new	Semantics.factorsOnImage_iff_no_split (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:40 Semantics.FiniteSplitCheck (def) formalization/lean/FoundationsVII/Semantics/Join.lean:63 Semantics.finiteSplitCheck_correct (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:67
Definition 4.4	new	Semantics.StrictObservable (def) formalization/lean/FoundationsVII/Semantics/Join.lean:93
Theorem 4.5	new	Semantics.strict_iff_split_pairs (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:99
Example 4.6	new	Semantics.cubeJoin (def) formalization/lean/FoundationsVII/Semantics/Join.lean:107 Semantics.cubeObservable (def) formalization/lean/FoundationsVII/Semantics/Join.lean:110 Semantics.cube_strict (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:119 Semantics.cube_first_not_strict (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:124
Definition 4.7	new	Semantics.CommonRefinement (def) formalization/lean/FoundationsVII/Semantics/Join.lean:79
Proposition 4.8	new	Semantics.factors_through_pair_of_refinement (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:83
Remark 4.9	new	Semantics.identity_refinement_recovers_cube (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:131
Definition 4.10	new	Semantics.RelabelOnly (def) formalization/lean/FoundationsVII/Semantics/Join.lean:141 Semantics.SchedulingOnly (def) formalization/lean/FoundationsVII/Semantics/Join.lean:146 Semantics.CoarseningOnly (def) formalization/lean/FoundationsVII/Semantics/Join.lean:150
Proposition 4.11	new	Semantics.split_composition (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:157 Semantics.strict_not_relabel (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:163 Semantics.strict_not_coarsening (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:171
Theorem 4.12	VII-C009 (semantic)	Semantics.SemanticStrictEvidence (structure) formalization/lean/FoundationsVII/Semantics/Join.lean:190 Semantics.semanticFlag (def) formalization/lean/FoundationsVII/Semantics/Join.lean:181 Semantics.strictCertificate (def) formalization/lean/FoundationsVII/Semantics/Join.lean:201 Semantics.strictCertificate_valid (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:222 Semantics.cubeCertificate_valid (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:244
Definition 5.1	new	Semantics.Star (inductive) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:7 Semantics.Joinable (def) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:24 Semantics.LocallyConfluent (def) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:27 Semantics.Confluent (def) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:30 Semantics.Terminates (def) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:33 Semantics.Normal (def) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:36
Theorem 5.2	new	Semantics.newman (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:59
Corollary 5.3	new	Semantics.normal_star_eq (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:63 Semantics.normal_forms_unique (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:69 Semantics.unique_reachable_normal (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:87

Item	Catalog	Lean declarations
Proposition 5.4	new	Semantics.finiteReachCheck_correct (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:130 Semantics.finiteJoinCheck_correct (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:194 Semantics.finitePeakCheckComplete_correct (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:220 Semantics.finitePeakCheckFin_correct (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:230
Remark 5.5	new	paper-only algorithmic observation; no Lean declaration
Example 5.6	new	Semantics.fourStep (def) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:238 Semantics.four_local (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:247 Semantics.four_not_confluent (theorem) formalization/lean/FoundationsVII/Semantics/Rewriting.lean:258
Definition 5.7	new	Semantics.AdmissionRules (structure) formalization/lean/FoundationsVII/Semantics/Admission.lean:7 Semantics.AdmissionRules.Enabled (def) formalization/lean/FoundationsVII/Semantics/Admission.lean:15 Semantics.AdmissionRules.Step (def) formalization/lean/FoundationsVII/Semantics/Admission.lean:18
Theorem 5.8	VII-C018 (semantic)	Semantics.AdmissionRules.terminates (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:32 Semantics.AdmissionRules.commuting_admissions (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:49 Semantics.AdmissionRules.locallyConfluent (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:62 Semantics.AdmissionRules.confluent (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:69 Semantics.AdmissionRules.unique_fixed_point (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:72 Semantics.AdmissionRules.star_preserves_nodup (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:26
Example 5.9	new	Semantics.disablingStep (def) formalization/lean/FoundationsVII/Semantics/Admission.lean:104 Semantics.disabling_terminates (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:107 Semantics.disabling_not_local (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:116 Semantics.disabling_two_fixed_points (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:152
Example 5.10	new	Semantics.freeRules (def) formalization/lean/FoundationsVII/Semantics/Admission.lean:87 Semantics.free_example (theorem) formalization/lean/FoundationsVII/Semantics/Admission.lean:90
Proposition 7.1	VII-C034 (semantic)	Semantics.lowerStep (def) formalization/lean/FoundationsVII/Semantics/Enablement.lean:7 Semantics.reachableVector (def) formalization/lean/FoundationsVII/Semantics/Enablement.lean:10 Semantics.form (def) formalization/lean/FoundationsVII/Semantics/Enablement.lean:39 Semantics.upperStep (def) formalization/lean/FoundationsVII/Semantics/Enablement.lean:42 Semantics.form_injective (theorem) formalization/lean/FoundationsVII/Semantics/Enablement.lean:51 Semantics.upper_step_iff_lower (theorem) formalization/lean/FoundationsVII/Semantics/Enablement.lean:65 Semantics.three_distinct_upper_states (theorem) formalization/lean/FoundationsVII/Semantics/Enablement.lean:74 Semantics.nontrivial_upper_formation (theorem) formalization/lean/FoundationsVII/Semantics/Enablement.lean:90 Semantics.enabled_not_descended (theorem) formalization/lean/FoundationsVII/Semantics/Enablement.lean:106 Semantics.observable_descends_of_fibre_constant (theorem) formalization/lean/FoundationsVII/Semantics/Enablement.lean:116

Item	Catalog	Lean declarations
Proposition 7.2	VII-C035 (semantic)	Semantics.EventSystem (structure) formalization/lean/FoundationsVII/Semantics/Precedence.lean:9 Semantics.Lawful (inductive) formalization/lean/FoundationsVII/Semantics/Precedence.lean:14 Semantics.sole_producer_join_absent (theorem) formalization/lean/FoundationsVII/Semantics/Precedence.lean:23 Semantics.external_source_allows_join (theorem) formalization/lean/FoundationsVII/Semantics/Precedence.lean:45
Proposition 7.3	VII-C015 (semantic)	Semantics.select (def) formalization/lean/FoundationsVII/Semantics/Selection.lean:10 Semantics.SelectionStep (def) formalization/lean/FoundationsVII/Semantics/Selection.lean:13 Semantics.selection_run_preserves (theorem) formalization/lean/FoundationsVII/Semantics/Selection.lean:21 Semantics.insertion_not_selection (theorem) formalization/lean/FoundationsVII/Semantics/Selection.lean:35
Proposition 7.4	VII-C021, VII-C033 (semantic)	Semantics.ReachWithin (inductive) formalization/lean/FoundationsVII/Semantics/Search.lean:8 Semantics.detect (def) formalization/lean/FoundationsVII/Semantics/Search.lean:17 Semantics.detector_sound (theorem) formalization/lean/FoundationsVII/Semantics/Search.lean:21 Semantics.detector_coverage (theorem) formalization/lean/FoundationsVII/Semantics/Search.lean:31 Semantics.beyond_horizon_null (theorem) formalization/lean/FoundationsVII/Semantics/Search.lean:43 Semantics.no_contact_with_one_use (theorem) formalization/lean/FoundationsVII/Semantics/ChannelSearch.lean:50 Semantics.contact_with_two_uses (theorem) formalization/lean/FoundationsVII/Semantics/ChannelSearch.lean:62
Example 6.3	VII-C009 (semantic)	Semantics.strictCertificate (def) formalization/lean/FoundationsVII/Semantics/Join.lean:201 Semantics.strictCertificate_valid (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:222 Semantics.cubeCertificate_valid (theorem) formalization/lean/FoundationsVII/Semantics/Join.lean:244
Definition 6.1	VII-C001	DomainState (structure) formalization/lean/FoundationsVII/Core/Domain.lean:13 DomainState.Coherent (def) formalization/lean/FoundationsVII/Core/Domain.lean:30 DomainState.normalForm_eq (theorem) formalization/lean/FoundationsVII/Access/Domain.lean:63
Definition S3.1	VII-C016	ContactSurface (structure) formalization/lean/FoundationsVII/Core/Contact.lean:13 ContactWitness (structure) formalization/lean/FoundationsVII/Core/Contact.lean:39 ContactMode (inductive) formalization/lean/FoundationsVII/Contact/Transport.lean:23 ContactChannelEvidence.rendezvous_licensed (theorem) formalization/lean/FoundationsVII/Contact/Transport.lean:60
Definition S3.2	VII-C008	JoinStatus (inductive) formalization/lean/FoundationsVII/Join/Status.lean:13 JoinAssessment.classify (def) formalization/lean/FoundationsVII/Join/Status.lean:56 Models.Finite.Phase3.status_partition_complete (theorem) formalization/lean/FoundationsVII/Models/Finite/Phase3/ContactJoinEnvelope.lean:161
Definition 6.2	VII-C009	StrictJoinEvidence.Certified (def) formalization/lean/FoundationsVII/Join/Strictness.lean:96 AntiProductWitness.Valid (def) formalization/lean/FoundationsVII/Join/Strictness.lean:28 StrictJoinEvidence.certified_has_anti_product_novelty (theorem) formalization/lean/FoundationsVII/Join/Strictness.lean:118 AntiProductWitness.valid_is_nonfactorizing (theorem) formalization/lean/FoundationsVII/Join/Strictness.lean:44

Item	Catalog	Lean declarations
Definition S3.3	VII-C012, VII-C013	EnablementRecord (structure) formalization/lean/FoundationsVII/Core/Enablement.lean:13 AttributedEnablement.WellFormed (def) formalization/lean/FoundationsVII/Enablement/Attribution.lean:77 EndogenousEnablementProfile.Eligible (def) formalization/lean/FoundationsVII/Enablement/Endogenous.lean:30 EndogenousEnablementProfile.eligible_iff_exact_criterion (theorem) formalization/lean/FoundationsVII/Enablement/Endogenous.lean:46
Definition S3.4	VII-C017	RouteComparison.Holonomy (def) formalization/lean/FoundationsVII/Dynamics/Holonomy.lean:60 constructive_interaction_holonomy (theorem) formalization/lean/FoundationsVII/Dynamics/Holonomy.lean:139 holonomy_with_zero_arrow_witness (theorem) formalization/lean/FoundationsVII/Dynamics/Holonomy.lean:142
Definition S3.5	VII-C032	DriveCertificate.Valid (def) formalization/lean/FoundationsVII/Dynamics/Holonomy.lean:80 ArrowClaim.Eligible (def) formalization/lean/FoundationsVII/Dynamics/Holonomy.lean:101 driven_arrow_requires_independent_drive_certificate (theorem) formalization/lean/FoundationsVII/Dynamics/Holonomy.lean:150 CrossTimeContact.Valid (def) formalization/lean/FoundationsVII/Dynamics/CrossTime.lean:37
Proposition 7.5	VII-C003, NGVII-01	AdmissionRegime.no_first_extension_without_seed_or_reachable_generator (theorem) formalization/lean/FoundationsVII/Access/Bootstrap.lean:63 NoGo.NGVII_01_escape_admitted_seed (theorem) formalization/lean/FoundationsVII/NoGo/Admission.lean:24 NoGo.NGVII_01_escape_reachable_generator (theorem) formalization/lean/FoundationsVII/NoGo/Admission.lean:31 NoGo.NGVII_01_escape_open_external_provision (theorem) formalization/lean/FoundationsVII/NoGo/Admission.lean:39
Remark 7.6	VII-C030	RefinementEffect (inductive) formalization/lean/FoundationsVII/Join/Retention.lean:102 no_unconditional_join_monotonicity_under_refinement (theorem) formalization/lean/FoundationsVII/Join/Retention.lean:256 refinement_effects_have_all_four_controls (theorem) formalization/lean/FoundationsVII/Join/Retention.lean:237
Remark 7.7	VII-C031	ContactQuantityTransformation.Conserved (def) formalization/lean/FoundationsVII/Join/Degree.lean:47 plausible_contact_quantities_disagree (theorem) formalization/lean/FoundationsVII/Join/Degree.lean:112 omitted_observer_occupancy_invalidates_conservation_credit (theorem) formalization/lean/FoundationsVII/Join/Degree.lean:119 exact_scoped_contact_conservation_is_available (theorem) formalization/lean/FoundationsVII/Join/Degree.lean:125
Remark 7.8	VII-C026	UniversalConstructionCertificate.RepresentsStrictJoin (def) formalization/lean/FoundationsVII/Join/Categorical.lean:116 mere_product_is_not_strict_join_credit (theorem) formalization/lean/FoundationsVII/Join/Categorical.lean:188 special_case_categorical_representation_is_conditional (theorem) formalization/lean/FoundationsVII/Join/Categorical.lean:199
Remark 7.9	VII-C028	AlgebraReadiness.Ready (def) formalization/lean/FoundationsVII/Dynamics/PrimitiveAlgebra.lean:25 full_generators_relations_program_deferred_with_formal_reopen_conditio n (theorem) formalization/lean/FoundationsVII/Dynamics/PrimitiveAlgebra.lean:70 resource_delta_fragment_is_associative (theorem) formalization/lean/FoundationsVII/Dynamics/PrimitiveAlgebra.lean:55

S3 The certificate records

This section defines the certificate records that the main paper summarizes in its section on the certificate language (main paper, Section 6). The strict-join certificate is defined there.

S3.1 Access and admission

The access record is defined in the main paper (Definition 6.1).

Admission is modelled as a typed transition system with five operations: admission, expiry, revocation, retraction and rollback. Each requires source, budget, guard and audit witnesses and has a typed effect on the admissibility coordinate. The four non-admitting transitions matter as much as admission: a system that can only admit cannot distinguish a claim that was withdrawn from one that was never made. An admission regime records whether its family of extensions is closed, which seeds it has admitted, which generators are reachable and which extensions are provisioned from outside. A first extension is *authorized* when there is an admitted seed, a reachable generator or open external provision; Proposition 7.5 is the resulting refusal.

S3.2 Contact and join

Definition S3.1 (Contact surface and contact witness). A *contact surface* records two theories, their interfaces, a crossing relation and an admissibility flag; it is well formed when the two theories differ and the crossing relation is named. A *contact witness* records a surface, an event, a source, whether the crossing occurred, a description of the payload and an audit; it is well formed when its surface is well formed and admissible, the crossing occurred, the payload is described and the audit is non-empty. Separately, channel evidence licenses one of five contact modes (rendezvous, mediation, sequential, shared carrier, shared budget) by a direct endpoint match, a certified mediator, a recorded sequence, a certified shared carrier or a certified shared budget.

Contact is weaker than join. Two theories may share a carrier, exchange a record or transport a payload without any joint theory coming into existence.

Definition S3.2 (Join status). Within a declared finite family of interfaces, a join attempt receives one of seven statuses: no evidenced contact, evidenced contact, common refinement, lawful composite, strict join, typed obstruction, or certified non-interaction.

The classifier assigns exactly one status to every attempt in the declared family, which is checked by enumeration over its carriers. Nothing is claimed about attempts outside the declared family.

The join budget prices interface access, parent retention, joint novelty, observer and instrument occupancy, and cross-costs, with refunds. An entry is credited when its amount is covered by payments and refunds, or when its amount is zero on a certified zero-cost channel. A formation earns *native* credit, credit as a product of the join's own sources rather than of bridged or external ones, only when every observer's occupancy is charged in full. The ledger is local accounting over declared resources, not a conservation law.

S3.3 Enablement and descent

Definition S3.3 (Enablement record). An *enablement record* identifies an operation made available, its source, whether it was executed, the target before and after, budgets, an attribution and an audit. An *attributed* enablement adds attribution evidence (a typed source, a witnessed execution, accounted budget and audit, and no hidden executor), requires execution, and requires the record's

source to match the evidence. Enablement is *endogenous* when the enabling generator is carried, reachable, executed, audited and budgeted within a closed boundary, neither a theorist nor an observer executor is hidden, and any environmental input is accounted.

An *enablement separation profile* stores independent Boolean flags. Two of them record claims: that an enablement is *load-bearing*, that is, the record asserts that the enabled operation depends on it; and that *descent* holds, that is, the record asserts that the enabled content descends from the enabling theory, which then also determines it. No semantic definition of dependence or descent is formalized; the flags are recorded assertions. Table S2 contains a profile whose load-bearing flag is set and whose descent flag is clear.

A transmission record carries a direction (upward, downward or peer), a payload, what was lost, what is ambiguous and a source. An operational profile records whether a rule is sound, executable, guard-active, reachable, fired and occurrent; it is coherent when an executable rule is sound, a reachable one is guard-active and executable, a fired one is reachable and an occurrent one has fired. A negative-evidence record states the force of a negative result: a point null, a bounded search, an exhaustive search of a finite family, and an unrestricted universal claim have four different forces, and the record states which one is claimed.

S3.4 Dynamics and residue

Definition S3.4 (Route comparison and holonomy). A *route comparison* records two route results, each with a target, a list of members, order and bracketing codes, a predictive value, an audit value and a legality flag. It has *route residue* when both results are legal, they share target and members, their order or bracketing codes differ, and their predictive or audit values differ. The catalog calls residue with differing predictive values *interaction holonomy*. The name is used in a declared structural sense; no bundle, connection or geometric phase is involved [1].

Definition S3.5 (Drive certificate). A *drive certificate* records five flags: a drive, a path asymmetry, a failed reversal, budgeting and auditing. It is valid when all five are set. An *arrow claim* is eligible only with a valid drive certificate. Cross-time contact additionally requires a synchronization or partial-order witness.

The fluctuation theorem relates forward and reverse path probabilities to entropy production for stochastic, microscopically reversible dynamics [2]; it is a comparison for the drive and reversal fields, not an interpretation of them.

The residual ledger types what an interaction leaves undischarged: inaccessible, unadmitted, expired, failed and join-created residuals. Each entry may record a discharge route (ascent, peer join, refinement or declared termination), and the ledger is append-only, so no residual disappears silently.

S4 Record-level separations and definitional statements

For each separation the finite reference world (main paper, Section 8) contains a named configuration that has the first property and lacks the second. Table S2 lists them. These are separations between certificate fields: each configuration is a record in the reference world, not an observed event, and the separations show that the fields are independent as records.

Table S2. Separations exhibited in the reference world. Each row names a property that does not imply another and the configuration that shows it.

Does not imply	Record in the reference world
<i>Access and admission</i>	
sound rule $\nRightarrow$ reachable	an operational profile marked sound but not reachable
reachable $\nRightarrow$ occurrent	an operational profile marked reachable but not occurrent
bounded null $\nRightarrow$ universal negative	a negative-evidence record over a family marked open
total lens $\nRightarrow$ total on a partial target	a transfer record with a total source lens, a partial self-owned target and no certified adapter
retrospective $\nRightarrow$ prospective credit	a commitment use registered after its evidence
<i>Contact and join</i>	
common origin $\nRightarrow$ shared access	two access records with the same source, carrier and instrument and different access fields
shared carrier $\nRightarrow$ transport	a record with a certified shared carrier and no licensed transport
compatible interface $\nRightarrow$ contact	a record with a common interface and no contact witness
contact $\nRightarrow$ strict join	a contact record with no anti-product witness
lawful parents $\nRightarrow$ a join exists	records of two lawful parents with no common admissible package recorded
product or common refinement $\nRightarrow$ strict	a record whose distinction is marked as factoring through the declared baseline
relabelling $\nRightarrow$ strict content	a join record whose novelty is marked relabelling-only
rescheduling $\nRightarrow$ strict content	a join record whose novelty is marked scheduling-only
one lineage $\nRightarrow$ source independence	a gate record with behavioural non-factorization and a single lineage
composite $\nRightarrow$ parent retention	a composite record with one parent marked not recoverable
non-factorization $\nRightarrow$ objecthood	a record marked non-factorizing without certified objecthood
refinement $\nRightarrow$ join preserved	a well-formed refinement record labelled <i>destroys</i>
<i>Enablement and descent</i>	
enablement $\nRightarrow$ descent	a separation profile marked load-bearing with no descent recorded
necessary enabler $\nRightarrow$ sufficient	a profile marked necessary and not sufficient, with alternative determinants recorded
enablement $\nRightarrow$ endogenous	a profile whose enabling step is marked theorist-executed
<i>Dynamics and residue</i>	
route residue $\nRightarrow$ arrow	a route comparison with residue and no drive certificate
strict join $\nRightarrow$ arrow	a strict-join record with directionality not certified
same members $\nRightarrow$ same result	a route comparison with equal members, different order codes and different values
one admission system $\nRightarrow$ one fixed point	a seed-partition record whose two terminal-package fields are unequal
joining $\nRightarrow$ fewer obstructions	a ledger transition discharging an inherited residual and marking a new one as join-created
zero stated cost $\nRightarrow$ zero cost	a join record stated at zero cost with positive observer occupancy recorded
omitted observer $\nRightarrow$ endogenous credit	an occupancy record charged below its occupied amount

S4.1 Statements that follow from the definitions

The catalog entries below are definitions, or statements that follow in a line from the definition of the certificate they concern (main paper, Section 7.3).

- *Neutral seeds.* A seed certified as neutral authorizes a first extension, but so does any admitted seed: authorization uses membership in the list of seeds, not the neutrality certificate. Two refusals attach to neutrality: a seed stocked after the fact earns no neutral credit, and a system-generated source routed through an external label is not thereby external.
- *Endogenous enablement.* The endogeny criterion is the definition of endogeny (Definition S3.3) read as an equivalence. Not definitional is that the two hidden-executor clauses are independent of the others: the reference world contains a theorist-triggered and an observer-triggered profile, each satisfying every other condition.
- *Downward selection.* A transmission record is a *pure downward selection* when it inserts nothing from outside and every lower-layer fact present after the selection was present before it or was inserted. With insertion excluded, no lower-layer fact is created: a downward explanation may not populate the lower layer by describing it. Proposition 7.3 proves this for selection modelled as intersection.
- *Horizons.* Non-occurrence within a finite horizon licenses only horizon-qualified conclusions unless the search family is closed, the detector’s power is established and the null is complete beyond the horizon. These are fields of the negative-evidence record, so the statement fixes what that record requires; Proposition 7.4 proves the corresponding implications for an explicit search.
- *Certified non-interaction.* A claim that two theories do not interact is admitted only with its contact channels, detector power, budget, horizon and family closure declared, and it covers only the declared family.
- *Composition of enablement.* The resource deltas of enablement links (cost and residual debt) combine associatively with a two-sided identity, for all inputs. The composability conditions (matching source types, execution witnesses, budgets and residual obligations) are not needed for associativity; they govern whether a composed chain earns credit. A two-link chain whose result depends on the order of its links is exhibited.
- *Confluence at a declared scope.* A criterion on a list of critical-pair records (every legal pair is marked joinable and audit-equivalent) is equivalent to confluence at the declared scope. Both sides are predicates on the same records, so the equivalence is a reformulation and proves no path confluence; the semantic statement is Theorem 5.8.
- *Join status and the strict-join certificate.* Over each declared interface family the classifier assigns exactly one of the seven statuses of Definition S3.2, as checked by exhaustive enumeration of the declared carriers. A valid strict-join certificate has objecthood, retention of both parents and a valid anti-product witness, and dropping any one of these admits a configuration that the certificate must refuse. These statements project fields of the certificate; their semantic counterpart is Theorem 4.12.
- *Order residue.* A route comparison has residue by Definition S3.4. The reference world exhibits residue without a drive certificate, which is why residue does not supply an arrow.

- *The join-created residual.* The reference world contains a ledger transition that discharges one inherited residual and records one new residual marked as created at the composite interface. The mark is recorded, not derived from a model of the join: the ledger can represent this situation and must keep the two kinds apart.
- *Cross-family corollaries.* The catalog’s 21 corollaries are conjunctions and projections of these certificates across clusters: a certified configuration carries several certificates at once, and each is discharged on its own. They are listed in Supplement Section S6.

S5 Certificate fields

The main paper summarizes the certificates by cluster (main paper, Section 6). Table S3 lists each certificate with what it records and its validity predicate. Table S4 lists the Lean structure that carries each one, the number of its fields and their names, read from the Lean sources. Validity predicates are separate definitions next to each structure.

Table S3. The certificates of the catalog. “Valid when” states the validity predicate of the corresponding Lean structure; an audit is required only where listed. The definitions cited are in the main paper.

Certificate	What it records	Valid when
<i>Access and admission</i>		
Access record	theory, interface, scope, time; seven access coordinates; audit (Definition 6.1)	the implication spine holds and the audit is non-empty
Admission step	one of five operations (admit, expire, revoke, retract, roll back), a transition between access records, source, budget and history before and after	the transition is well formed, its guard holds, the source is authorized, the ledgers and history extend, the cost is covered, and the effect on admissibility is the typed one
Admission regime	closure of the family, admitted seeds, reachable generators, externally provisioned and permitted first extensions, audit	the four lists have no repetitions and the audit is non-empty
Neutral provisioning	source, prospective commitment, task-blindness, outcome independence, symmetry, control, audit	the source is external, accepted and not system-generated; the commitment is accepted, well formed, registered before the observation and active at it; the four flags are set; the audit is non-empty
Prospective commitment	registration, validity window, expiry, future transition, reserved budget, source and permitted source kinds, preregistration, disposition, audit	registration precedes the window, the window is ordered, permitted source kinds are listed, it is preregistered and the audit is non-empty
Bridge contract	source and target declarations, types and maps; hypotheses preserved, added and lost; trust dependencies; non-implications; disposition; audit	declarations, types and maps are named, non-implications are listed and the audit is non-empty; transport is licensed only by an accepted well-formed bridge, never by a citation
<i>Contact and join</i>		
Contact surface	two theories, their interfaces, a crossing relation, an admissibility flag (Definition S3.1)	the theories differ and the crossing relation is named
Contact witness	a surface, an event, a source, a crossing flag, a payload description, audit	the surface is well formed and admissible, the crossing occurred, the payload is described and the audit is non-empty

Certificate	What it records	Valid when
Join entry	parents, surface, contact state and optional witness, source and budget ledgers, observers, obstructions, status, audit	at least two parents, a well-formed surface, well-formed ledgers, a status consistent with the contact state, and a non-empty audit
Join status	one of seven statuses (Definition S3.2)	assigned by the classifier
Strict-join certificate	candidate, objecthood, retention of each parent, anti-product witness, source independence, budget settlement, directionality, audit (Definition 6.2)	the candidate is well formed, the objecthood, retention, independence and budget flags are set, the witness is valid and the audit is non-empty
Source-independence gate	the ancestries of the two sides, behavioural non-factorization, ancestry witnessed, honest bridge	for credit: independent ancestries, a witnessed ancestry, an honest bridge and behavioural non-factorization
Join cost entry and ledger	per entry: resource, kind, amount, paid, refunded, zero-cost certification, disposition; the ledger adds an audit	each entry has amount zero on a certified zero-cost channel, or amount at most paid plus refunded; the ledger has an entry and an audit
Observer occupancy	observer, source kind, visible interfaces, capacity, occupied and charged amounts, audit	occupancy within capacity, some interface visible, audit non-empty; native credit additionally needs charged equal to occupied
<i>Enablement and descent</i>		
Enablement record	operation made available, source, execution flag, target before and after, budgets, attribution, audit (Definition S3.3)	the operation is named, the target is one theory not moving back in time, the audit is non-empty; attributed enablement also needs valid attribution evidence, execution and a matching source
Transmission record	direction, source and target theories, payload, source, typing and budget flags, loss and ambiguity descriptions, lower-layer fact before and after, external insertion, causal certificate, audit	distinct theories, a named payload, a typed and budgeted source, described loss and ambiguity, and a non-empty audit
Operational profile	sound, executable, guard active, reachable, fired, occurrent	executable implies sound, reachable implies guard active and executable, fired implies reachable, occurrent implies fired
Negative evidence	grade, scope, carrier, family, family closure, search bound, detector power, escape routes, audit	carrier, family and detector power described, escape routes listed, a closed family for an exhaustive grade, and a non-empty audit
Coverage certificate	covered channels, exact coverage, family closure, detector power, budget, horizon, observed contact, escape routes, audit	channels listed, the five coverage flags set, no contact observed, escape routes listed, audit non-empty
<i>Dynamics and residue</i>		
Route comparison	two route results, each with target, members, order and bracketing codes, predictive and audit values and a legality flag (Definition S3.4)	residue: both legal, same target and members, a changed code, and a different predictive or audit value
Drive certificate	drive, path asymmetry, failed reversal, budgeted flag, audited flag (Definition S3.5)	all five flags set
Residual entry and ledger	per entry: kind, parents, description, activity, discharge route, cross-term and relabelling flags, audit; the ledger adds an audit	every entry has parents, a description and an audit, cross-terms have two parents and are not relabellings, dissolved entries are inactive; the ledger is non-empty and audited

Certificate	What it records	Valid when
Contact quantity	before and after vectors, a named scalar with its two values, observer, instrument and residual inclusion flags, equality check, audit	named scalar, all three coordinates included, equal values, equality checked, audit non-empty

Table S4. Certificate structures and their fields.

Structure	Fields	Field names
DomainState formalization/lean/FoundationsVII/ Core/Domain.lean:13	12	theoryId; interfaceId; scopeId; timestamp; expressible; present; exposed; recoverable; admissible; reachable; occurrent; audit
AdmissionRegime formalization/lean/FoundationsVII/ Access/Bootstrap.lean:14	6	closedFamily; admittedSeeds; reachableGenerators; externallyProvisionedExtensions; permittedFirstExtensions; audit
TypedAdmissionStep formalization/lean/FoundationsVII/ Access/Transitions.lean:38	8	operation; transition; sourceBefore; sourceAfter; budgetBefore; budgetAfter; historyBefore; historyAfter
NeutralProvisioningCertificate formalization/lean/FoundationsVII/ Access/Bootstrap.lean:118	8	certificateId; source; commitment; taskBlind; outcomeIndependent; symmetryCertified; controlMatched; audit
ProspectiveCommitment formalization/lean/FoundationsVII/ Core/Admission.lean:62	11	commitmentId; registeredAt; validFrom; expiresAt; futureTransition; reservedBudget; sourceId; permittedSourceKinds; preregistered; disposition; audit
BridgeContract formalization/lean/FoundationsVII/ Core/Protocols.lean:26	14	bridgeId; sourceDeclaration; targetDeclaration; sourceType; targetType; sourceMap; targetMap; preservedHypotheses; addedHypotheses; lostHypotheses; trustDependencies; nonclaims; disposition; audit
ContactSurface formalization/lean/FoundationsVII/ Core/Contact.lean:13	6	leftTheory; rightTheory; leftInterface; rightInterface; crossingRelation; admissible
JoinEntryRecord formalization/lean/FoundationsVII/ Join/Entry.lean:25	11	entryId; parents; surface; contactState; contact; sources; budgets; observers; obstructions; status; audit
StrictJoinEvidence formalization/lean/FoundationsVII/ Join/Strictness.lean:81	10	certificateId; candidate; objecthoodCertified; leftParentRecoverable; rightParentRecoverable; antiProduct; sourceIndependent; budgetSettled; directionalityCertified; audit
AntiProductWitness formalization/lean/FoundationsVII/ Join/Strictness.lean:13	10	jointDistinctionPresent; factorsThroughLeft; factorsThroughRight; factorsThroughDeclaredProduct; factorsThroughCommonRefinement; relabelOnly; schedulingOnly; coarseningOnly; witnessDescription; audit
SourceIndependenceGate formalization/lean/FoundationsVII/ Join/Source.lean:78	5	left; right; behavioralNonfactorization; ancestryWitnessed; bridgeHonest
JoinCostEntry formalization/lean/FoundationsVII/ Join/Budget.lean:20	7	resourceId; kind; amount; paid; refunded; zeroCostCertified; disposition
JoinPaymentLedger formalization/lean/FoundationsVII/ Join/Budget.lean:74	2	entries; audit
ObserverOccupancyRecord formalization/lean/FoundationsVII/ Core/Observer.lean:9	7	observerId; sourceKind; visibleInterfaces; capacity; occupied; charged; audit
EnablementRecord formalization/lean/FoundationsVII/ Core/Enablement.lean:13	9	enablementId; enabledOperation; sourceId; executed; targetBefore; targetAfter; budgets; attribution; audit
TransmissionRecord formalization/lean/FoundationsVII/ Dynamics/Transmission.lean:30	14	direction; sourceTheory; targetTheory; payload; sourceId; sourceTyped; budgeted; lossDescription; ambiguityDescription; lowerFactPresentBefore; lowerFactPresentAfter; externalInsertion; causalCertificate; audit

Structure	Fields	Field names
OperationalProfile formalization/lean/FoundationsVII/ Access/Reachability.lean:14	6	sound; executable; guardActive; reachable; fired; occurent
NegativeEvidenceRecord formalization/lean/FoundationsVII/ Core/Protocols.lean:224	10	recordId; grade; scope; carrierDescription; familyDescription; familyClosed; searchBound; detectorPowerDescription; escapeRoutes; audit
ContactCoverageCertificate formalization/lean/FoundationsVII/ Join/NonInteraction.lean:77	10	certificateId; coveredChannels; exactCoverage; familyClosed; detectorPower; budgetSufficient; horizonComplete; observedContact; escapeRoutes; audit
RouteComparison formalization/lean/FoundationsVII/ Dynamics/Holonomy.lean:25	2	left; right
DriveCertificate formalization/lean/FoundationsVII/ Dynamics/Holonomy.lean:70	5	drivePresent; pathAsymmetry; reversalFails; budgeted; audited
ResidualEntry formalization/lean/FoundationsVII/ Residuals/Ledger.lean:30	9	residualId; kind; sourceParents; sourceDescription; active; dischargeRoute; crossTerm; relabelOnly; audit
ResidualLedger formalization/lean/FoundationsVII/ Residuals/Ledger.lean:60	2	entries; audit
ContactQuantityTransformation formalization/lean/FoundationsVII/ Join/Degree.lean:32	10	before; after; declaredScalarName; beforeScalar; afterScalar; observerIncluded; instrumentIncluded; residualIncluded; exactEqualityChecked; audit

S6 Register of the catalog

S6.1 The 36 entries

Table S5 gives each entry its identifier and name, its kind (Section 3.2), where the main paper discusses it, what it establishes, and the evidence attached to it.

Table S5. Register of the catalog.

Entry	Kind; place	What is established	Evidence
VII-C001 Accessible-domain state normal form	definition Definition 6.1	Defines the seven-coordinate access record and its coherence spine. Each failed converse has a coherent witness in the reference world.	assays P5-E11; countermodels CM-08, CM-09, CM-12; 48 Lean declarations
VII-C002 Lawful admission transition system	definition Section 6	Defines five typed admission transitions, each requiring source, budget, guard and audit witnesses, with a typed effect on admissibility.	assays P5-E01; countermodels CM-08, CM-09, CM-18; 41 Lean declarations
VII-C003 Bootstrap obstruction theorem	consequence of definitions (Proposition 7.5) Proposition 7.5	In a closed regime with no admitted seed and no reachable generator no first extension is authorized; this follows from the definition of authorization. Each of the three escape routes is realized by a regime record.	assays P5-E06; countermodels CM-19, CM-20; 31 Lean declarations
VII-C004 Neutral seed and provisioning certificate	definition; authorization does not use neutrality Section 7.3	Defines the neutral-provisioning certificate and refuses neutral credit to retrospectively stocked or relabelled system-generated seeds. Authorization of a first extension does not use neutrality.	countermodels CM-01, CM-19; 44 Lean declarations
VII-C005 Common-origin non-transfer law	separations and a refusal Section 7.1	Exhibits access records with the same origin, carrier and instrument and different access fields, and a record with shared access and a common source; a shared lineage fails the independence gate by definition.	countermodels CM-01, CM-12, CM-26; 35 Lean declarations

Entry	Kind; place	What is established	Evidence
VII-C006 Prospective commitment certificate	definition Section 6	Defines the prospective-commitment record; a use registered after its evidence cannot discharge the prospective certificate.	assays P5-E01; countermodels CM-20, CM-09; 26 Lean declarations
VII-C007 Join-entry record normal form	definition Section 6	Defines the join-entry record. A well-formed record does not establish that a join exists.	assays P3-E01, P3-E02; countermodels CM-02, CM-03, CM-14; 32 Lean declarations
VII-C008 Join existence and obstruction status calculus	definition Definition S3.2	Defines seven join statuses; over each declared finite interface family the classifier assigns exactly one, checked by enumeration.	assays P3-E02, P5-E03, P5-E11; countermodels CM-02, CM-13, CM-27; 40 Lean declarations
VII-C009 Strict join certificate and anti-product witness	definition; semantic counterpart Theorem 4.12 Definition 6.2	Defines the strict-join certificate; its necessity clauses are projections of recorded fields. Semantic counterpart: main paper, Theorem 4.5 and Theorem 4.12, which cover the parents and their pairing but not every common refinement (Remark 4.9).	assays P3-E03, P3-E10, P5-E01, P5-E02, P5-E07; countermodels CM-03, CM-13, CM-14, CM-16; 44 Lean declarations
VII-C010 Source-independence gate	definition Section 6	Defines the source-independence gate; behavioural non-factorization alone does not earn independence-sensitive credit.	assays P3-E04, P5-E02; countermodels CM-01, CM-26; 28 Lean declarations
VII-C011 Join budget and payment ledger	definition Section 6	Defines the join ledger; an entry is credited when payments plus refunds cover its amount, or when its amount is zero on a certified zero-cost channel.	assays P3-E05, P3-E06, P5-E01, P5-E02, P5-E06; countermodels CM-11, CM-21; 32 Lean declarations
VII-C012 Enablement record and attribution calculus	definition Definition S3.3	Defines the enablement record with its attribution field; a hidden executor defeats attribution.	assays P4-E01, P5-E11; countermodels CM-06, CM-07, CM-17; 39 Lean declarations
VII-C013 Endogenous enablement criterion	definition read as an equivalence Section 7.3	The endogeny criterion is the definition read as an equivalence. The two hidden-executor controls each satisfy every other condition.	assays P4-E02, P5-E09; countermodels CM-11, CM-17; 35 Lean declarations
VII-C014 Birth/contact-surface classification	definition Section 6	Defines the birth record; a contact record can be marked relation-only.	assays P4-E03; countermodels CM-02, CM-17; 42 Lean declarations
VII-C015 Typed transmission and descent-fidelity law	consequence of the definition; semantic counterpart Proposition 7.3 Section 7.3	Defines transmission records; for a pure downward selection the conclusion that no lower fact is created is contained in the definition.	assays P4-E04, P5-E05; countermodels CM-06, CM-22, CM-23; 46 Lean declarations
VII-C016 Peer contact and transport without join	separation (existence of transport without a composite) Section 7.1	Exhibits a well-formed peer-transport record marked as forming no composite.	assays P3-E01; countermodels CM-02, CM-19; 42 Lean declarations
VII-C017 Interaction-order residue and holonomy	definition; separation from the arrow Definition S3.4	Defines route residue for two route results; exhibits a comparison record with residue and no arrow credit.	assays P4-E09, P4-E10, P5-E07, P5-E10; countermodels CM-10, CM-24; 46 Lean declarations

Entry	Kind; place	What is established	Evidence
VII-C018 Admission confluence and seed-dependence law	reformulation; semantic counterpart Theorem 5.8 Section 7.3	Relates two predicates on the same critical-pair records; this is a reformulation and proves no path confluence. Exhibits a seed-partition record whose two terminal-package fields are unequal. Semantic counterpart: main paper, Theorem 5.8, which gives a unique terminal state from each state when admission only enables, and Example 5.9.	assays P4-E07, P4-E08, P5-E10; countermodels CM-23, CM-24; 39 Lean declarations
VII-C019 Access/join residual and obstruction-dissolution ledger	definition Section 6	Defines the append-only residual ledger with inherited, dissolved and join-created classes; scope changes do not delete residuals silently.	assays P3-E08, P5-E04, P5-E11; countermodels CM-22, CM-25; 25 Lean declarations
VII-C020 Bridge and semantic-withdrawal discipline	definition Section 6	Defines the bridge contract; a citation alone does not license transport.	countermodels CM-12, CM-18; 10 Lean declarations
VII-C021 Reachability, guard activity, and horizon law	certificate definition; semantic counterpart Proposition 7.4 Section 7.3	Defines the operational profile and the horizon-qualified negative certificate; reachability alone does not give occurrence.	assays P5-E08; countermodels CM-08, CM-09, CM-18; 45 Lean declarations
VII-C022 Exposure, recoverability, admissibility, and rigidity calculus	definition Section 6	Defines interface coordinates for determination, exposure, recoverability and admissibility; equal determination can have different exposure.	countermodels CM-03, CM-12; 48 Lean declarations
VII-C023 Negative-result quantifier discipline	definition Section 6	Defines the negative-evidence record; a point null does not license an unrestricted negative.	assays P5-E08; countermodels CM-18, CM-27; 5 Lean declarations
VII-C024 Claim-grade and normative-specification discipline	definition Section 3.2	Defines the grade record; finite evidence does not upgrade a claim's grade.	countermodels CM-12, CM-18; 10 Lean declarations
VII-C025 Two-Theory World detector contract	decidable finite statement Section 8	Over the 24 scenarios and 27 named countermodels the detector contract is well formed and complete and every expected status is decided; checked by evaluation.	assays P5-E11; countermodels CM-01, CM-02, CM-03, CM-04, CM-05, CM-06, CM-07, CM-08, CM-09, CM-10, CM-11, CM-12, CM-13, CM-14, CM-15, CM-16, CM-17, CM-18, CM-19, CM-20, CM-21, CM-22, CM-23, CM-24, CM-25, CM-26, CM-27; 8 Lean declarations
VII-C026 Categorical reduction decision	deferred; checklist only Remark 7.8	Deferred. The categorical certificate is a checklist of flags and supports no categorical statement; a mere product earns no strict-join credit.	assays P3-E10; countermodels CM-13, CM-27; 21 Lean declarations
VII-C027 Enablement-chain composition law	associativity unconditional; credit needs composability Section 7.3	Resource deltas combine associatively with an identity for all inputs; composability governs credit. Exhibits a two-link chain record whose composed values differ with order.	assays P4-E06, P5-E04; countermodels CM-07, CM-24; 45 Lean declarations
VII-C028 Primitive-operation algebra readiness criterion	deferred; monoid of resource deltas Remark 7.9	Deferred; the resource-delta fragment is a monoid.	assays P4-E12; countermodels CM-24; 26 Lean declarations

Entry	Kind; place	What is established	Evidence
VII-C029 Observer and instrument occupancy law	refusal Section 7.4	Native or endogenous credit is refused while charged occupancy is below actual occupancy; external observer and zero occupancy are escapes.	assays P5-E02, P5-E09; countermodels CM-11, CM-21; 29 Lean declarations
VII-C030 Parent refinement, retention, and join descent	refuted; four-way classification Remark 7.6	Refuted at the level of records: a well-formed refinement record labelled destroys exists, so unconditional preservation is not permitted. All four effect labels occur among the records of the declared carrier; no model of refinement acting on a join is given.	assays P3-E07, P5-E05; countermodels CM-05, CM-22; 29 Lean declarations
VII-C031 Contact-degree conservation decision	two scalars disagree Remark 7.7	Two particular scalar degrees disagree on one vector, and an apparent conservation omitting observer occupancy fails the accounting certificate. Nothing universal is proved.	assays P3-E11; countermodels CM-21, CM-25; 22 Lean declarations
VII-C032 Interaction arrow, irreversibility, and cross-time contact	refusal without a drive certificate Definition S3.5	An arrow claim requires a drive certificate; exhibits a comparison record with residue and no arrow credit. Cross-time contact requires a synchronization or partial-order witness.	assays P4-E10, P4-E11, P5-E07, P5-E10; countermodels CM-10, CM-24; 58 Lean declarations
VII-C033 Coverage-qualified certified non-interaction	certificate definition; semantic counterpart Proposition 7.4 Section 7.3	Defines the coverage certificate for non-interaction; a declared exhaustive family excludes contact within that family only.	assays P3-E09, P5-E03; countermodels CM-02, CM-18, CM-27; 39 Lean declarations
VII-C034 Enablement without descent separation theorem	separation profiles Section 7.1	Exhibits Boolean separation profiles: load-bearing enablement with no recorded descent, and a necessary but insufficient enabler.	assays P4-E05, P5-E05; countermodels CM-06, CM-07; 25 Lean declarations
VII-C035 No-free-join and self-bootstrap no-go family	refusals; first clause Proposition 7.2 Section 7.4	Payment and relabelling refusals follow from the ledger and anti-product definitions. The first clause, that a join cannot create its own first capability, is Proposition 7.2 of the main paper.	assays P3-E05, P5-E06; countermodels CM-03, CM-19, CM-21; 53 Lean declarations
VII-C036 Join-created obstruction and needle law	recorded attribution Section 7.3	The ledger can record a transition that discharges one inherited residual and marks one new residual as join-created. The mark is recorded, not derived from a model of a join.	assays P3-E08, P5-E04; countermodels CM-25, CM-03; 25 Lean declarations

S6.2 Recorded non-implications

Table S6 lists, for each entry, the non-implications recorded with it and its escape routes.

Table S6. Recorded non-implications and escape routes.

Entry	Does not claim	Escape routes
VII-C001	Domain width is not capability. No total order among all status coordinates is claimed. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C002	A well-formed rule need not be executable. A reachable transition need not fire. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Entry	Does not claim	Escape routes
VII-C003	External provision is not forbidden. The statement is scoped to the declared transition family. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C004	Neutrality is audit-relative, not metaphysical independence. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C005	Common origin may still be relevant evidence when correctly typed. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C006	Commitment does not guarantee reachability or success. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C007	Record completeness does not prove a join exists. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C008	The taxonomy is not universal without a completeness theorem. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C009	Strict join does not imply an arrow. Parent strictness alone does not prove joint strictness. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C010	Source independence is not sufficient for strict novelty or causal independence. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C011	No universal conserved interaction currency is asserted. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C012	Enablement does not imply descent or sufficient determination. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C013	Endogenous does not mean uncaused or environmentally isolated. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C014	Contact does not itself constitute birth. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C015	Construction direction is not automatically temporal directionality. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Entry	Does not claim	Escape routes
VII-C016	Peer transport is not joint objecthood or strict novelty. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C017	P3 holonomy does not imply an arrow. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C018	One commuting square does not prove global confluence. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C019	Access growth need not monotonically reduce all obstructions. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C020	A bridge certificate does not strengthen its source theorem. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C021	Non-occurrence alone is not impossibility. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C022	Exposure is not determination; recoverability is not permission. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C023	A failed instance never refutes an unrestricted existential claim. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C024	Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C025	Passing finite scenarios does not prove universal laws. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C026	No unconditional categorical reduction is claimed. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C027	Enablement is not assumed transitive without hypotheses. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C028	No complete primitive algebra is claimed. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

Entry	Does not claim	Escape routes
VII-C029	Observation is not prohibited; unrecorded load-bearing observation is. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C030	Retention alone is not strict novelty. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C031	No conserved contact quantity is claimed. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C032	Order sensitivity and entropy production internal to a parent do not alone prove an inter-theory arrow. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C033	No finite null universalizes beyond its covered family. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C034	The theorem does not deny that some enablement also descends. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C035	The family permits admitted seeds, declared bridges, and proven zero-cost channels. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.
VII-C036	Joining is not monotonically obstruction-reducing. Finite reference-world success does not universalize the statement.	Narrow or change the declared carrier/interface family. Add an explicit source, seed, bridge, budget, or drive witness omitted by the no-go hypotheses.

S6.3 Refusals

Table S7 lists the eleven refusals of the main paper’s Table 3 with the entries they concern and the Lean theorems that exhibit their escape routes.

Table S7. Refusals and escape theorems.

Identifier	Refusal	Entries	Escape theorems
NGVII-01	No first extension without seed or reachable generator	VII-C003	NGVII_01_escape_admitted_seed NGVII_01_escape_reachable_generator NGVII_01_escape_open_external_provisio n
NGVII-02	No resemblance-only independence-sensitive join	VII-C010, VII-C035	NGVII_02_escape_witnessed_independent_ contact
NGVII-03	No retrospective self-certification	VII-C006	NGVII_03_escape_prospective_certificat e
NGVII-04	No automatic total-lens transfer	VII-C005	NGVII_04_escape_certified_adapter
NGVII-05	No unpriced observer-native formation credit	VII-C029	NGVII_05_escape_external_observer NGVII_05_escape_zero_occupancy
NGVII-06	No join from contact alone	VII-C007, VII-C009, VII-C035	NGVII_06_escape_complete_strict_join_c ertificate

Identifier	Refusal	Entries	Escape theorems
NGVII-07	No product/common-refinement strictness credit	VII-C009, VII-C026, VII-C035	NGVII_07_escape_anti_product_witness
NGVII-08	No one-lineage source-independence credit	VII-C010, VII-C035	NGVII_08_escape_disjoint_root_certificate
NGVII-09	No positive-cost join credit without payment; a zero-cost join on a certified zero-cost channel is credited without payment	VII-C011, VII-C035	NGVII_09_escape_paid_channel NGVII_09_escape_certified_zero_cost_channel
NGVII-10	No arrow from holonomy alone	VII-C017, VII-C032	NGVII_10_escape_independent_driven_arrow
NGVII-11	No occurrence from reachability alone	VII-C021	NGVII_11_escape_fired_occurrence

S6.4 Cross-family corollaries

The 21 corollaries are conjunctions and projections of certificates from different clusters (main paper, Section 7.3). Table S8 lists each with its Lean theorem, the entries it spans and its location.

Table S8. Cross-family corollaries.

Identifier	Lean theorem	Entries	Location
FVII-COR-001	certified_has_temporal_source_join_and_payment	C002, C006, C009, C011	formalization/lean/FoundationsVII/Corollaries/ProspectiveJoin.lean:40
FVII-COR-002	retrospective_registration_blocks_prospective_join_credit	C002, C006, C009, C011	formalization/lean/FoundationsVII/Corollaries/ProspectiveJoin.lean:59
FVII-COR-003	certified_preserves_declared_source_alignment	C002, C006, C009, C011	formalization/lean/FoundationsVII/Corollaries/ProspectiveJoin.lean:70
FVII-COR-004	certified_has_independence_payment_and_capacity_bound	C009, C010, C011, C029	formalization/lean/FoundationsVII/Corollaries/SourceBudget.lean:24
FVII-COR-005	source_and_payment_gates_remain_distinct	C009, C010, C011, C029	formalization/lean/FoundationsVII/Corollaries/SourceBudget.lean:39
FVII-COR-006	certified_noninteraction_excludes_strict_join	C008, C033	formalization/lean/FoundationsVII/Corollaries/NonInteraction.lean:19
FVII-COR-007	certified_noninteraction_retains_escape_routes	C008, C033	formalization/lean/FoundationsVII/Corollaries/NonInteraction.lean:35
FVII-COR-008	certified_accumulates_resources_and_preserves_residuals	C019, C027, C036	formalization/lean/FoundationsVII/Corollaries/EnablementResidual.lean:28
FVII-COR-009	join_created_needle_blocks_zero_residual_reading	C019, C027, C036	formalization/lean/FoundationsVII/Corollaries/EnablementResidual.lean:44
FVII-COR-010	preserving_refinement_with_valid_descent_preserves_three_gates	C015, C030, C034	formalization/lean/FoundationsVII/Corollaries/RefinementDescent.lean:22
FVII-COR-011	destroying_refinement_exhibits_a_failed_join_gate	C015, C030, C034	formalization/lean/FoundationsVII/Corollaries/RefinementDescent.lean:36
FVII-COR-012	valid_descent_does_not_erase_refinement_obstruction	C015, C030, C034	formalization/lean/FoundationsVII/Corollaries/RefinementDescent.lean:45
FVII-COR-013	no_free_first_join_two_gate_obstruction	C003, C011, C035	formalization/lean/FoundationsVII/Corollaries/NoFreeJoin.lean:12

Identifier	Lean theorem	Entries	Location
FVII-COR-014	seeded_and_paid_first_join_escape	C003, C011, C035	formalization/lean/FoundationsVII/Corollaries/NoFreeJoin.lean:29
FVII-COR-015	strict_join_and_holonomy_do_not_replace_drive	C009, C017, C032	formalization/lean/FoundationsVII/Corollaries/HolonomyArrow.lean:12
FVII-COR-016	holonomy_with_independent_drive_supports_arrow	C009, C017, C032	formalization/lean/FoundationsVII/Corollaries/HolonomyArrow.lean:26
FVII-COR-017	certified_excludes_contact_and_occurrence_at_declared_scope	C021, C023, C033	formalization/lean/FoundationsVII/Corollaries/NegativeForce.lean:23
FVII-COR-018	point_null_plus_bounded_horizon_is_not_automatically_global	C021, C023, C033	formalization/lean/FoundationsVII/Corollaries/NegativeForce.lean:36
FVII-COR-019	certified_requires_pricing_and_internal_execution	C013, C029	formalization/lean/FoundationsVII/Corollaries/ObserverEndogeny.lean:22
FVII-COR-020	unpriced_observer_blocks_joint_native_endogenous_credit	C013, C029	formalization/lean/FoundationsVII/Corollaries/ObserverEndogeny.lean:38
FVII-COR-021	zero_occupancy_escape_preserves_endogenous_credit	C013, C029	formalization/lean/FoundationsVII/Corollaries/ObserverEndogeny.lean:48

S7 Finite evidence

S7.1 Scenarios, countermodels and the detector contract

Twenty-four scenarios and twenty-seven named countermodels are declared against the reference world (main paper, Section 8). A scenario exercises a certificate and carries the status it is expected to produce; a countermodel is built to refute one named inference and carries the reason it is rejected. The separations of Table S2 are the twenty-seven countermodels. A detector contract records append-only evidence, non-empty lists of signal, null and falsifier cases, non-empty lists of controls of four kinds (same source, no contact, rescheduling and relabelling) and a false-positive cost above the false-negative cost. Over the 24 scenarios and 27 countermodels the contract is well formed and complete, and every scenario’s and countermodel’s expected status is decided by the finite evaluator; this is a decidable statement checked by evaluation. Canonical counts below are smaller than raw counts only where a declared symmetry, such as exchanging the two parents or renaming sources, is a genuine symmetry of the property checked; the accepted and rejected counts record what each carrier contains and are not evidence for or against a law.

S7.2 The 43 assays

Table S10 lists every assay with its raw, canonical and accepted cardinalities and the family it enumerates. The totals are 284,524 raw, 214,636 canonical, 70,915 accepted and 143,721 rejected cases. Each count holds for its declared carrier only.

Table S9 groups the assays by cluster.

The largest single assay enumerates omissions from the strict-join certificate: 131,072 raw cases, reducing to 65,536 canonical cases under exchange of the parents, of which 32,770 satisfy the certificate. Most other assays are two or three orders of magnitude smaller, because the property checked has fewer degrees of freedom.

Table S9. Finite assays by group. Each is exhaustive over its declared carrier only.

Group	Assays	Raw	Canonical	Accepted
Access and admission	9	1,332	1,076	470
Contact and join	11	141,112	73,528	34,532
Enablement and dynamics	12	55,168	55,168	14,832
Integration across clusters	11	86,912	84,864	21,081
Total	43	284,524	214,636	70,915

Table S10. Finite assays.

Assay	Raw	Canon.	Acc.	Family
<i>Access and admission</i>				
P2-E01	128	128	14	Seven-coordinate access normal forms satisfying only the declared implication spine.
P2-E02	128	128	12	Operational profiles separating rule soundness, execution, reachability, firing, and occurrence.
P2-E03	32	32	29	Bootstrap assignments with admitted-seed, reachable-generator, and open-family external-provision authorization.
P2-E04	256	256	129	Prospective-credit assignments with preregistration, precedence, task blindness, outcome independence, symmetry, and matched controls.
P2-E05	512	256	128	Two-package provenance/access profiles normalized by source-ID renaming.
P2-E06	32	32	19	Total-lens transfer assignments with self-owned targets and explicit partial-target adapter licensing.
P2-E07	64	64	33	Finite-horizon negative-force assignments with family closure and detector-power gates.
P2-E08	18	18	15	Observer occupancy/pricing assignments over costs 0,1,2.
P2-E09	162	162	91	Failed-admission settlement assignments over costs 0,1,2.
<i>Contact and join</i>				
P3-E01	256	256	54	Compatibility, evidenced contact, peer transport, composite formation, and strict-join claims.
P3-E02	4,096	2,048	170	Declared-family join evidence profiles, canonicalized under left/right parent exchange.
P3-E03	131,072	65,536	32,770	Strict-join omission attacks over independently represented certificate axes.
P3-E04	128	128	72	Source-lineage and independence-sensitive credit assignments.
P3-E05	432	432	210	Typed join payment, refund, observer occupancy, and certified-zero-cost assignments.
P3-E06	200	200	100	Finite live-join capacity profiles over capacities and positive unit costs up to four.
P3-E07	576	576	138	Parent-retention and refinement effects with preserving, strengthening, weakening, and destroying cases.
P3-E08	192	192	11	Inherited, dissolved, and join-created residual assignments with typed cross-term provenance.
P3-E09	256	256	128	Coverage-qualified non-interaction profiles with an explicit outside-family escape marker.
P3-E10	1,024	1,024	99	Conditional product, pullback, and pushout representations of strict join.
P3-E11	2,880	2,880	780	Candidate contact/join quantities over bounded values with exact scoped conservation credit.
<i>Enablement and dynamics</i>				
P4-E01	1,792	1,792	917	Source-typed enablement attribution profiles with honest bridge-refinement stability.
P4-E02	2,048	2,048	1,024	Exact carried/reachable/executed/audited/budgeted endogenous-credit profiles.
P4-E03	1,280	1,280	160	Closure-agency and relation/layer/participant-birth profiles with objecthood gates.
P4-E04	6,144	6,144	1,605	Upward/downward/peer transmission profiles with payload, loss, ambiguity, source, budget, and causal-certificate gates.

Assay	Raw	Canon.	Acc.	Family
P4-E05	512	512	128	Constructive enablement/descent and necessary/sufficient separation profiles.
P4-E06	41,472	41,472	10,449	Two-link enablement composition with source/budget/residual/audit compatibility and accumulated cost/debt coordinates.
P4-E07	64	64	32	Finite critical-pair profiles distinguishing joinability, audit equivalence, terminal equality, and predictive equivalence.
P4-E08	64	64	21	Seed-partition dependence profiles separated from presentation-only route differences.
P4-E09	1,024	1,024	256	Typed order/bracketing comparison profiles defining route residue and predictive holonomy on one target.
P4-E10	128	128	64	Holonomy/drive/path-asymmetry/reversal profiles with an independent arrow certificate.
P4-E11	384	384	96	Cross-time contact via synchronization or partial-order witnesses with admissible reparameterization controls.
P4-E12	256	256	80	Primitive-operation algebra readiness profiles plus a separately scoped resource-delta fragment.
<i>Integration across clusters</i>				
P5-E01	8,192	8,192	4,096	Prospective commitment, lawful admission, strict-join, source, and payment integration.
P5-E02	3,456	3,456	864	Source-independent, observer-priced strict-join credit and finite live-capacity assignments.
P5-E03	512	512	255	Coverage-qualified certified non-interaction and strict-join exclusion.
P5-E04	18,432	18,432	4,826	Residual-aware two-link enablement composition with append-only retention and cross-term debt.
P5-E05	2,048	2,048	496	Parent-refinement effects composed with source-, budget-, and square-preserving descent fidelity.
P5-E06	4,096	4,096	256	Combined bootstrap and payment assignments for first-join credit.
P5-E07	1,024	1,024	400	Strict-join, route-holonomy, and independently driven arrow assignments.
P5-E08	4,096	4,096	288	Operational reachability plus horizon-, closure-, and detector-qualified negative force.
P5-E09	18,432	18,432	2,304	Observer-pricing and declared-boundary endogeneity assignments.
P5-E10	8,192	6,144	1,152	Three-domain critical-pair, seed-partition, and interaction-order assignments.
P5-E11	18,432	18,432	6,144	Integrated two- and three-domain release profiles with typed sources, budgets, observers, joins, residuals, and scoped negatives.

S7.3 The named countermodels

Table S11 lists the 27 named countermodels of the reference world: the inference each refutes, and the scenario it is declared against. The scenario column names the reference scenario the countermodel is attached to; it is not a status of the countermodel itself.

Table S11. Named countermodels.

Id	Name	Shows	Scenario
CM-01	Same source but no shared access	Common origin does not imply shared access or independence.	TTW-S05
CM-02	Common interface without witnessed contact	Compatibility does not imply contact or non-interaction.	TTW-S12
CM-03	Contact without strict joint information	Contact does not imply strict join.	TTW-S06
CM-04	Scheduling-only apparent joint novelty	Post-hoc scheduling can manufacture an apparent joint signal.	TTW-S14
CM-05	Join that loses one child	Composite formation does not imply parent retention.	TTW-S09
CM-06	Enablement without descent	An enabling condition can be load-bearing without factorization of the enabled object.	TTW-S03

Id	Name	Shows	Scenario
CM-07	Enablement necessary but insufficient	An enabler may be necessary while alternatives still determine the result.	TTW-S03
CM-08	Sound but unreachable admission rule	Soundness does not imply operational reachability.	TTW-S19
CM-09	Reachable admission that never occurs	Reachability does not imply occurrence.	TTW-S20
CM-10	Holonomy with zero arrow	Route residue does not imply directionality.	TTW-S17
CM-11	Observer occupancy omitted from budget	Hidden observer resources defeat endogenous/native credit.	TTW-S22
CM-12	Total-lens result fails on partial self-owned access	Totality assumptions do not transfer silently.	TTW-S23
CM-13	Product/common refinement without strictness	Common refinement is not anti-product novelty.	TTW-S07
CM-14	Relabeling-only fake join	Presentation change cannot manufacture strict content.	TTW-S15
CM-15	Strict extension without certified objecthood	Nonfactorization does not imply closure/objecthood.	TTW-S15
CM-16	Strict join without directionality	Strict novelty does not imply a P6 arrow.	TTW-S08
CM-17	Theorist-triggered enablement is not endogenous	Execution by the analyst fails the carried-generator condition.	TTW-S02
CM-18	Finite negative search with open family	A bounded null does not prove a universal no-go.	TTW-S19
CM-19	Shared carrier without lawful peer transport	Co-location does not provide an executable bridge or first capability.	TTW-S01
CM-20	Post-hoc prediction self-certification	Retrospective admission cannot earn prospective credit.	TTW-S04
CM-21	Zero-cost join hides positive observer cost	Unpriced occupancy invalidates no-cost credit.	TTW-S22
CM-22	Parent refinement destroys an admissible join	Join admissibility need not be monotone under parent refinement.	TTW-S24
CM-23	Seed-dependent closure fixed points	Initial partition/seed can change the terminal package.	TTW-S24
CM-24	Identical members assembled in different orders	Bracketing/order may change the audited result.	TTW-S16
CM-25	Join manufactures a cross-term needle	Joining need not monotonically dissolve obstructions.	TTW-S24
CM-26	One-lineage pseudo-join	Behavioral difference inside one lineage does not establish source independence.	TTW-S05
CM-27	Lawful parents with no common admissible package	Parent lawfulness does not guarantee join existence.	TTW-S13

S8 Formal status

S8.1 The certificate library

The certificate library has 1,477 public declarations: 732 theorems and 745 definitions, structures, inductive types and abbreviations. For every theorem an executed `#print axioms` report was recorded.

Axioms	Theorems	Share
none	625	85.4%
propext	97	13.3%
propext, Quot.sound	10	1.4%

No theorem depends on `Classical.choice` or on `sorryAx`. The library is built on `leanprover/lean4:v4.28.0` with no external library, together with byte-identical copies of

the Foundations V and VI libraries and of the earlier series' formal cores. The structure used for commuting descent squares takes its four carriers in Lean's lowest universe `Type`; every other general construction is universe-polymorphic.

S8.2 The semantic modules

The three semantic modules have 45 theorems, including four private helpers. Their axiom reports are listed in Table S12. Nine use no axioms, five use `propext` only, four use `propext` and `Quot.sound`, and twenty-seven use `propext`, `Classical.choice` and `Quot.sound`. The modules contain no `sorry`, no added axiom, no `admit` and no `native_decide`. The finite reachability search enumerates all sublists of a covering list and is exponential in its length.

Table S12. Axiom reports of the semantic modules.

Theorem	Axioms
<code>Star.single</code>	none
<code>Star.trans</code>	none
<code>newman</code>	none
<code>normal_star_eq</code>	none
<code>normal_forms_unique</code>	none
<code>unique_reachable_normal</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>filter_mem_allSubsets</code>	<code>propext</code> , <code>Quot.sound</code>
<code>finiteClosedCheck_correct</code>	<code>propext</code> , <code>Quot.sound</code>
<code>finiteReachCheck_correct</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>finitePeakCheck_correct</code>	<code>propext</code> , <code>Quot.sound</code>
<code>finiteJoinCheck_correct</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>finitePeakCheckComplete_correct</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>finitePeakCheckFin_correct</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>four_local</code>	<code>propext</code>
<code>four_not_confluent</code>	<code>propext</code>
<code>AdmissionRules.step_preserves_nodup</code>	<code>propext</code>
<code>AdmissionRules.star_preserves_nodup</code>	<code>propext</code>
<code>AdmissionRules.terminates</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>AdmissionRules.commuting_admissions</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>AdmissionRules.locallyConfluent</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>AdmissionRules.confluent</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>AdmissionRules.unique_fixed_point</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>free_example</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>disabling_terminates</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>disabling_not_local</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>disabling_two_fixed_points</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>factors_implies_onImage</code>	none
<code>factorsOnImage_iff_no_split</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>finiteSplitCheck_correct</code>	<code>propext</code> , <code>Quot.sound</code>
<code>factors_through_pair_of_refinement</code>	<code>propext</code>
<code>strict_iff_split_pairs</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>cube_strict</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>cube_first_not_strict</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>identity_refinement_recovers_cube</code>	none
<code>split_composition</code>	none
<code>strict_not_relabel</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>strict_not_coarsening</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>semanticFlag_true</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>semanticFlag_false</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>strictCertificate_valid</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>cubeCertificate_valid</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>join_of_stars</code>	none
<code>reaches_normal</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>AdmissionRules.enabled_after_other</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>cubeEvidence</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>selection_step_preserves</code>	none
<code>selection_run_preserves</code>	none
<code>insertion_creates_fact</code>	none

Theorem	Axioms
insertion_not_selection	none
sole_producer_join_absent	propext
external_source_allows_join	none
external_source_join_occurs	propext
formation_enabled	none
enabled_observable_not_descended	propext, Classical.choice, Quot.sound
coarse_refinement_loses_cube	propext, Classical.choice, Quot.sound
fine_refinement_retains_cube	none
detector_sound	propext, Quot.sound
detector_coverage	propext, Quot.sound
beyond_horizon_null	propext, Quot.sound
no_contact_in_covered_family	propext, Quot.sound
zero_horizon_coverage	propext
zero_horizon_null	none
later_witness	none
one_horizon_sound	none
one_horizon_detects_contact	none
detected_contact_witness	propext, Quot.sound
all_bool_covered	propext
stuck_reach_only_false	none
stuck_closed_at_zero	none
concrete_no_contact	propext, Quot.sound
lower_reachable_iff	propext, Quot.sound
lower_supplies_upper	none
upper_requires_lower	none
form_injective	propext, Quot.sound
form_surjective	none
upper_step_iff_lower	propext, Quot.sound
three_distinct_upper_states	propext, Quot.sound
nontrivial_upper_formation	propext, Quot.sound
quotient_commutates_with_formation	propext
enabled_not_descended	propext, Classical.choice, Quot.sound
observable_descends_of_fibre_constant	propext, Classical.choice, Quot.sound
quotient_observable_descends	propext, Classical.choice, Quot.sound
zero_channel_reach_eq	none
one_channel_coverage	propext
one_channel_negative_scan	none
no_contact_with_one_use	propext, Quot.sound
two_channel_uses_reach_contact	none
contact_with_two_uses	none

S8.3 What the formal development does not establish

The kernel checks that each encoded statement follows from its encoded hypotheses. It does not check that a certificate is the right record of an interaction, that a declared finite carrier is the relevant one, or that a result is significant. Many theorems of the certificate library are short consequences of the definitions of their records (main paper, Section 7.3). Two statements of the main paper are argued on paper only: the elementary remark on $a \oplus b$ and the first clause of the no-free-join refusal (main paper, Section 10).

S9 Replay

All commands run from the repository root.

Lean library, including the semantic modules.

```
cd formalization/lean
lake build FoundationsVII
```

Finite evidence, registries and reports.

```
bash scripts/rebuild_fvii_sci05.sh
bash scripts/run_fvii_sci05_external_lean.sh
```

The first command regenerates the finite evidence from the Python reference implementation and rebuilds the registries and reports. The second runs the Lean kernel build, the axiom reports and the comparisons between the Lean and Python implementations.

Supplement tables and papers.

```
python3 paper/supplement/build_supp.py
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build main.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build supplement.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build main.tex
```

The main paper and the supplement cite each other's numbered items, so the main paper is built a second time after the supplement.

References

- [1] Michael V. Berry. Quantal Phase Factors Accompanying Adiabatic Changes. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 392(1802):45–57, 1984. doi:10.1098/rspa.1984.0023.
- [2] Gavin E. Crooks. Entropy Production Fluctuation Theorem and the Nonequilibrium Work Relation for Free Energy Differences. *Physical Review E*, 60(3):2721–2726, 1999. doi:10.1103/PhysRevE.60.2721.
- [3] Ioannis Tsiokos. Six Birds: Foundations of Emergence Calculus, 2026. Six Birds Theory, Foundations I.
- [4] Ioannis Tsiokos. Six Birds Foundations II: Admissibility, Meta-Theory, and the Exact-Six Program, 2026. Six Birds Theory, Foundations II.
- [5] Ioannis Tsiokos. Six Birds Foundations III: A Finite Audited Interaction Calculus for SBT, 2026. Six Birds Theory, Foundations III.
- [6] Ioannis Tsiokos. Six Birds Foundations IV: A Catalog of Layer-Agnostic Structural Laws, 2026. Six Birds Theory, Foundations IV.
- [7] Ioannis Tsiokos. Six Birds Foundations V: Endogenous Closure — A Catalog of Structural Laws for Living, Cognitive, and Social Systems, 2026. Six Birds Theory, Foundations V.
- [8] Ioannis Tsiokos. Six Birds Foundations VI: A Catalog of Dynamical Structural Laws, 2026. Six Birds Theory, Foundations VI.